Wajeeh Ahmed

AI/ML Researcher | Computer Science Undergraduate | Technical Instructor

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PROFESSIONAL SUMMARY

Computer Science undergraduate and AI/ML researcher with 4 Springer-published papers on machine learning and deep learning for intrusion detection systems, including a Best Paper Award at ICETI 2025. Hands-on experience building ML/DL models (CNN, TensorFlow, scikit-learn) and full-stack applications (FastAPI, React, PostgreSQL, PHP/MySQL). Experienced technical instructor who has delivered Cisco CCNA training to 350+ students across 5+ universities and served as a university Teaching Assistant. Seeking an AI/ML or research role to solve real-world problems.

EDUCATION

Bachelor of Science in Computer Science — Present
Shaheed Zulfikar Ali Bhutto Institute of Science and Technology (SZABIST) — Islamabad, Pakistan

- **Relevant Coursework:** Programming, Data Structures & Algorithms, Databases, Software Engineering, Operating Systems, Computer Networks, Research Methods

Intermediate (High School) — Aug 2023
Bahria College Islamabad — Islamabad, Pakistan

AWARDS & RECOGNITION

- **Best Paper Award** and **Best Presentation Award**, ICETI 2025 (Springer)
- **Best Poster Award**, COSIM, Germany
- Recognized as the youngest researcher at SZABIST to achieve 4 international publications
- Certificate of Appreciation, NAVTTC, for exceptional contribution to student training

WORK EXPERIENCE

Research Assistant (Higher Education) — Oct 2025 – Present
AI-Eyes (Explain You Science) — Islamabad, Pakistan

- Conduct applied AI research and develop solutions to complex problems under the mentorship of Dr. Inamullah Khan.
- Coordinate end-to-end planning and execution of international conferences, managing registrations, scheduling, and on-site operations for 350+ participants.

Web Developer — Aug 2025 – Present
AI-Eyes (Explain You Science) — Remote

- Revamped the AI-Eyes website, architecting and integrating an internship portal, student portal, and international research paper submission system.
- Maintain the platform and ship feature enhancements that improve user experience and functionality.

Teaching Assistant – Computer Networks & Data Communication (CNDC) — Jun 2026 – Aug 2026
SZABIST, under Mr. Salman — Islamabad, Pakistan

- Supported course instruction for the CNDC course, assisting students with networking concepts, lab work, and assignments.
- Helped with grading, held student help sessions, and supported the instructor in course delivery and classroom management.

Vocational Instructor – CCNA — Aug 2025
National Vocational and Technical Training Commission (NAVTTC) — Islamabad, Pakistan

- Delivered instructor-led Cisco CCNA networking course to undergraduate students under NAVTTC's national skills development program.
- Guided students through networking fundamentals, hands-on labs, and Cisco certification requirements, achieving successful certification outcomes.

Computer Science Instructor – CCNA Aug 2025

AI-Eyes (Explain You Science) Islamabad, Pakistan

- Delivered Cisco's official CCNA course to 350+ students across 5+ universities through hybrid onsite and online instruction.
- Prepared students for the Cisco CCNA Certified Technician certification through structured training and hands-on lab exercises.

Web Developer Intern Jun 2025 – Sep 2025

Young-Dev (Jun – Sep 2025) · Arch Technologies (Jul – Aug 2025) Remote

- Built responsive front-end websites, landing pages, and client pages using HTML and CSS, collaborating with teams to deliver to client requirements.

PROJECTS

Tax Saathi – Smart Tax Filing Assistant for Pakistani Salaried Workers

Final Year Project (team of 3) · Python, FastAPI, PostgreSQL, React, TypeScript, Tailwind CSS, Flutter, Docker

- Building a guided tax-filing platform that turns salaried users' withholding and deduction data into exact figures for the FBR IRIS portal, in English and Urdu, removing the need for paid tax agents.
- Designing a rule-based tax engine driven by versioned rule files (Tax Years 2026–2027), with planned document scanning of salary/deduction certificates and import of previous returns to cut manual entry.
- Developed a FastAPI REST backend with JWT authentication, email OTP verification, two-factor login, password reset, and role-based admin and taxpayer access.
- Secured sensitive data with bcrypt password hashing and Fernet encryption of CNIC numbers at rest; built consent management, identity-data deletion, and tax-year preparation tracking.
- Containerized PostgreSQL and a Mailpit test inbox with Docker Compose; backend covered by pytest integration tests.

AI-Eyes Website Re-architecture

AI-Eyes (Explain You Science) · HTML, CSS, JavaScript

- Rebuilt a fragile legacy site that had no entry point (no index.html), a separate hand-made HTML page per team member, and images scattered outside any assets folder.
- Re-architected it into a clean, maintainable structure with a proper index, reusable page templates, and an organized assets directory; prototyped a "Liquid Glass" UI, then delivered the final design to the approved reference (pending sign-off).

Online Exam Management System (OEMS)

HTML, CSS, PHP, MySQL · Awarded full marks

- Built a full-stack web app where students view exam datesheets and pass/fail results by roll number, replacing notice boards and manual queries.
- Developed a secure admin panel to create and update datesheets (subject, date, time, venue) and student results, with all data stored in a MySQL database via PHP.

DSA LIB – Header-Only C++ Data Structures Library

C++17, Templates, Smart Pointers, G++

- Built a type-safe, template-based, header-only library covering linked lists (singly/doubly/circular), stacks, queues, binary trees, BST, AVL, heaps, hash tables, and graphs (BFS/DFS).
- Implemented merge, quick, heap, and insertion sort plus linear and binary search, with iterator support and exception handling; validated edge cases and performance with custom test drivers.

OOP-Based Banking System with CSV Storage

C++, OOP, File I/O

- Built a console banking app supporting customer profiles, multiple Checking/Current accounts, password-verified login, deposits, and withdrawals.
- Applied inheritance, abstraction (virtual functions), composition, and encapsulation; persisted customers and accounts (IBAN, type, balance, creation date) to CSV files, parsed with ifstream/stringstream and exception handling.

RESEARCH PROJECTS & PUBLICATIONS

Comparative Study on Optimized Models for Intrusion Detection System Using the CICIDS2017 Dataset Oct 2025

First Author · Springer · Python, scikit-learn, TensorFlow

- Compared and optimized ML/DL models on CICIDS2017 for higher detection accuracy and efficiency, outperforming standard baselines; built the full cleaning, training, and evaluation pipeline.

Anomaly-Based Intrusion Detection System for UAVs Using the CICIoV2024 Dataset Oct 2025
Co-Author · Springer

- Developed an anomaly-based IDS framework for UAV networks against real-time cyber threats; handled preprocessing, feature engineering, training, and result analysis.

Lightweight Intrusion Detection System for UAV-Based Search and Rescue Operations Using Machine and Deep Learning Models Oct 2025
Co-Author · Springer

- Proposed a lightweight ML/DL IDS improving reliability and energy efficiency in resource-constrained UAVs; conducted data cleaning, training, and benchmarking.

A Lightweight Anomaly-Based Intrusion Detection System for UAV Networks Using Hybrid Deep Learning 2026
Co-Author · Springer, ICETI 2025 (accepted, to appear)

- Designed a hybrid deep learning model that improves UAV intrusion detection accuracy at low computational cost.

UPCOMING PUBLICATIONS

Beyond the Data Center: Profiling AI Sequence Architectures on Edge Unified Memory
First Author · COSIM, Germany · Best Poster Award

- Profiled AI sequence model architectures on edge devices with unified memory to evaluate their performance outside data-center hardware.

A Scalable Ensemble Framework for Handling Severe Class Imbalance in Next-Gen Cloud Intrusion Detection Systems
First Author · 2nd ICITIST, Baku, Azerbaijan

- Built an XGBoost + LightGBM ensemble with SMOTE-Tomek balancing on CSE-CIC-IDS2018, detecting rare cloud attacks with a false positive rate of just 0.42%.

Evaluating Deep Learning Architectures for Intrusion Detection and Forensic Analytics in Industrial Cyber-Physical Systems
First Author · 2nd ICITIST, Baku, Azerbaijan

- Applied sequence-aware LSTM models on the GothX industrial framework to detect slow-moving anomalies across IIoT subsystems.

SKILLS

Programming Languages: Python, C++ (C++17), TypeScript, JavaScript, PHP, SQL, Dart
Machine Learning & AI: TensorFlow, scikit-learn, NumPy, Pandas, NLTK, Matplotlib, Deep Learning (CNN), Anomaly Detection, Feature Engineering, Model Evaluation
Backend & Databases: FastAPI, REST APIs, JWT Authentication, PostgreSQL, MySQL, SQLAlchemy
Tools & Platforms: Git, GitHub, GitLab, Docker, Docker Compose, pytest
Frontend & Mobile: React, Tailwind CSS, HTML, CSS, Responsive Design, Flutter
Networking: Cisco CCNA, Computer Networks
Research: Academic Writing, Quantitative & Qualitative Research, Literature Review
Teaching: Instructional Strategies, Virtual Learning Environments, CCNA Certification Training
Core CS: Object-Oriented Programming (OOP), Data Structures & Algorithms
Languages: English (Proficient, C2), Urdu (Native)

CERTIFICATIONS

- Introduction to Cybersecurity – Cisco (May 2026) · AI Tools – IBM (Sep 2025)
- Introduction to AI – Google (Aug 2025) · Maximize Productivity with AI Tools – Google (Aug 2025)
- Front-End Web Development with HTML & CSS – UC Davis, Coursera (Jul 2025)

LEADERSHIP & VOLUNTEERING

Student Ambassador SZABIST, Islamabad
Microsoft Learn Student Ambassadors

- Organized and led workshops and events on Microsoft technologies and career development; also an active member of Google Developer Student Clubs (GDSC).

Blood Donation Volunteer Islamabad, Pakistan
Red Crescent Society

- Donated blood, mobilized volunteers for donation drives, and helped organize and promote community health initiatives.